MIL-C-20/16D
25 May 1973
SUPERSEDING
MIL-C-0020/16C(USAF)
15 December 1972, and
MIL-C-20/16A
2 June 1961

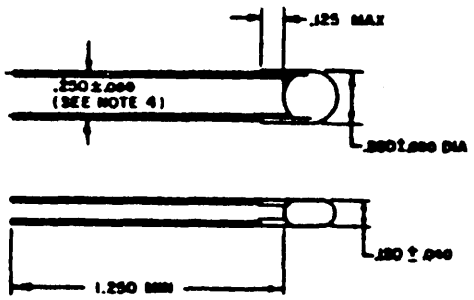
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING) (INSULATED), NON-ESTABLISHED RELIABILITY, STYLE CC50

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION. SEE MIL-C-20/43 FOR ITEMS PREVIOUSLY COVERED IN MIL-C-0020/16C(USAF).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-20.



INCHES	MM
.000	.76
.000	1.52
.000	1.52
.120	3.05
.125	3.18
.000	6.35
.000	7.11
1.000	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned wire of .020±.002(.04±.05 mm) diameter.
4. This dimension shall be measured at the point of emergence of the leads from the body of the capacitor.

FIGURE 1. Style CC50 capacitors (insulated).

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Lead type: Radial.
DC rated voltage: 500 volts.
Operating temperature range: -55° to +125°C.
Characteristic: See table I.

Capacitance: Within tolerance specified (see table I).

Quality factor (Q): In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.
Body insulation - Test II.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Capacitance (pF)	Capacitance tolerance	Temperature coefficient shown as column head and tolerance envelope for temperature coefficient shown in column							
			C	H	L	P	R	S	T	U
CC30-J3R0-	3	CD	J	J						
CC30-J3R3-	3.3	CD	J	J						
CC30-H4R0-	4	CD	H	H	H	H				
CC30-H4R7-	4.7	CD	H	H	H	H				
CC30-H5R0-	5	CDF	H	H	H	H	H	H		
CC30-6R0-	6	CDF	H	H	H	H	H	H	J	
CC30-6R8-	6.8	CDF	H	H	H	H	H	H	J	
CC30-7R0-	7	CDF	H	H	H	H	H	H	J	
CC30-8R0-	8	CDF	H	H	H	H	H	H	J	J
CC30-9R0-	9	CDF	H	H	H	H	H	H	J	J
CC30-100-	10	CDFG	H	H	H	H	H	H	J	J
CC30-120-	12	GJK	.....	.....	H	H	H	H	J	J
CC30-150-	15	GJK	.....	.....	.....	.....	H	.....	J	J
CC30-J180-	18	GJK	.....	.....	.....	.....	.....	.....	J	J
CC30UJ200-	20	GJ	.....	.....	.....	.....	.....	.....	.....	J
CC30UJ220-	22	GJK	.....	.....	.....	.....	.....	.....	.....	J
CC30UJ240-	24	GJ	.....	.....	.....	.....	.....	.....	.....	J
CC30UJ270-	27	FGJK	.....	.....	.....	.....	.....	.....	.....	J

1/ Complete type designation will include additional letter symbols to indicate, where applicable, temperature coefficient, tolerance envelope for temperature coefficient, and capacitance tolerance.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. For capacitors with values of 20 pF or less, the temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

Life: In accordance with MIL-C-20, operating condition 1.

Marking: Method I - In accordance with MIL-C-20, except for "JAN" brand.
Method II - In accordance with MIL-C-20 and as shown on figure 3.

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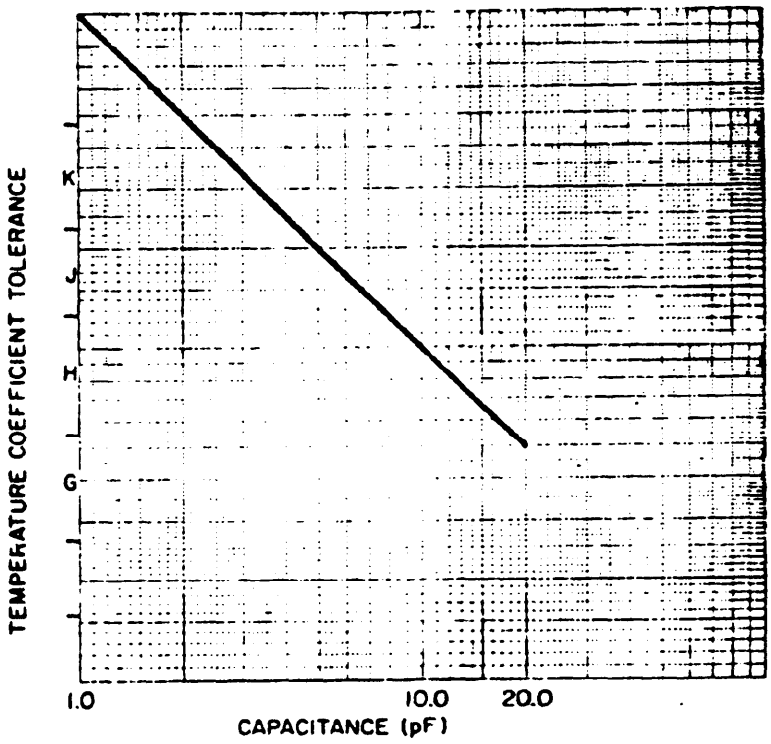
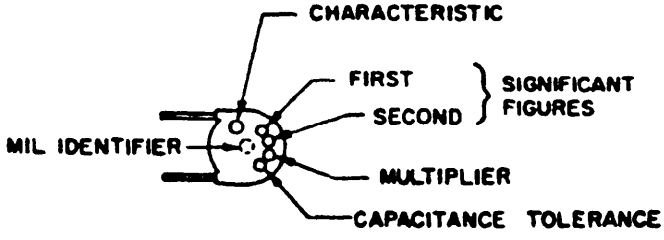


FIGURE 2. Limits for characteristic measurement.



1. The characteristic, capacitance, and capacitance tolerance shall be indicated on the capacitor by five spots of color. The color spots shall be applied so that the designation may be read clockwise. The diameter of the characteristic spot shall be at least twice that of any other spot, or the space between the first and last spots shall be at least three times the space between adjacent spots.

2. The MIL identifier (black spot) shall be applied to the capacitor body approximately in the center and on the reverse side of the capacitor body to which the color code is applied.

FIGURE 3. Color-code marking.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 80
DSA - ES

User activities:
Army -
Navy - AS, SH, MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-1064-16)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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